

RetroDiff: Training Objective Derivation

Let x denote a clean data sample and x_t its noised counterpart at diffusion step $t \in \{1, \dots, T\}$. The forward process $q(x_{1:T}|x)$ gradually corrupts x ; the model p_θ learns to reverse it. Applying Jensen’s inequality to the log-likelihood yields the *evidence lower bound* (ELBO):

$$\log p_\theta(x) \geq \mathbb{E}_{q(x_{1:T}|x)} \left[\log \frac{p_\theta(x_{0:T})}{q(x_{1:T}|x)} \right] = -\mathcal{L}_{\text{ELBO}} \quad (1)$$

Because the Markov structure of the forward process allows the joint to factorise, the bound in (1) decomposes into a reconstruction term at $t=1$ and a sum of per-step KL divergences that penalise each denoising transition:

$$\mathcal{L}_{\text{ELBO}} = \underbrace{\mathbb{E}_q[-\log p_\theta(x_0|x_1)]}_{\text{reconstruction}} + \sum_{t=2}^T \underbrace{\text{KL}(q(x_{t-1}|x_t, x) \parallel p_\theta(x_{t-1}|x_t))}_{\text{step-}t \text{ denoising}} \quad (2)$$

RetroDiff’s key contribution is to condition every denoising step on a set $R(x) = \{r_1, \dots, r_k\}$ of exemplars retrieved from the training corpus at inference time. Substituting the conditioned ELBO from (2) into an expectation over the data distribution gives the final per-parameter objective:

$$\mathcal{L}_{\text{RetroDiff}}(\theta) = \mathbb{E}_{x \sim p_{\text{data}}} [\mathcal{L}_{\text{ELBO}}(x | R(x))] \quad (3)$$

Minimising (3) over θ trains the reverse process to exploit retrieved neighbours, reducing the effective variance of each denoising step without altering the ELBO’s theoretical guarantees.